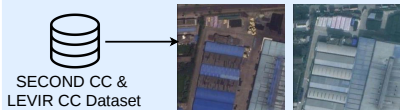


Caption Synthesize & Label Extraction



Captions

1. Building is built in a barren region between buildings.
2. Blue building is constructed in empty region between buildings.
3. More buildings are constructed around blue buildings.
4. Rectangle building built in a barren region around blue buildings.
5. In the space between buildings a blue rectangle building is built.

Label Normalization

Decomposition (Inter-class):

"Building built in barren region"

Object/Action Norm:

"Buildings are constructed"

Taxonomy & Change Ontology

Land cover:

1. Building
2. Bareland
3. Water
4. Vehicle
5. Vegetation
6. Road

Change type:

1. Added
2. Removed
3. Replaced

Semantic Mapping

Decomposition (Inter-class):

"Building built in barren region" → ["Building Added", "Barehand Removed"]

Object/Action Norm:

"Buildings are constructed" → ["Building Added"]

Quality Control & Validation

Rule-based algorithm + Manual audit

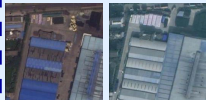
Compositional Triplet Generation

Subset Sampling

$$L(l_{ref}) \subset L(l_{target})$$

CTCR-RS Triplets

Reference Pair



Text Change:

Same changes, plus vegetation added.

Target Pair



Ground truth set:

{Visual Change + Text Change}